Contents

The Heisenberg Cut as a Physical Threshold: Location, Width, and Consequences of the Classical–Quantum Boundary in Detector Dynamics

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Contributions. Claude Fable 5 (Anthropic) performed the formalization, the manuscript prose, the simulation and figure, and the literature work. John M. Bramble, MD supplied the physical framing and research direction, adjudicated scope and interpretation, and is the accountable human sponsor. Byline order is the sponsor's decision and may be revised before any submission.

Abstract. The Heisenberg cut — the point at which the von Neumann measurement chain is severed and unitary description gives way to definite outcomes — is standardly a bookkeeping convention: movable, unphysical, chosen for convenience, and famously objectionable for exactly that reason. We argue that the cut is a physical threshold with a computable location and a computable width. The argument imports one theorem from a companion paper on outcome selection: below the registration threshold a driven absorber mode has no autonomous phase — it is a damped linear system, globally contracted onto a drive-slaved fixed point, with decay rate $\gamma = \Delta E/\hbar$ set by its off-shellness — while above threshold the mode owns a phase that can lock. The cut sits where the phase-restoring rate falls to the coupling, $\gamma \sim K$, and its fractional width is $w = K/\omega$: sharp beyond resolution for atomic systems ($w \sim 10^{-8}$–10^{-6}), percent-level for broadband solid-state absorbers ($w \sim 10^{-2}$). On this account the celebrated movability of the conventional cut is explained rather than denied — within the reversible sector the placement of the system–apparatus split is provably free, and that freedom hardens into physics only at the locking layer. The framework cleanly separates three notions that measurement talk chronically conflates — entangling interaction, decoherence, and the nonlinear lock — and locates every interpretive dispute in the third. Consequences follow for the anatomy of a measurement event (capture, selection, registration, with reversibility ending precisely at the layer), for the virtual/real distinction (off-shell excitation as a failed lock whose lifetime is the energy–time uncertainty; on-shell as locked, with critical slowing at the boundary), and for experiment: the cut has an address in every laboratory, the layer width is a dial that existing platforms already span, and the threshold picture makes coupling — not mass, size, or gravitational self-energy — the variable that decides where quantum

behavior ends, a discriminator against both spontaneous-collapse and gravitational-reduction programs. All claims are conditional on the companion paper's single-detector premises; nothing here consumes its nonlocal ontology.

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1. Introduction

Quantum mechanics describes a measurement twice. Inside the formalism, apparatus and system entangle unitarily and nothing ever happens; outside it, an experimenter reads one number off one dial. Somewhere between the two descriptions a line is drawn — system on one side, classical apparatus on the other — and the theory's predictions are indifferent to where. Heisenberg noted the indifference; von Neumann proved a version of it, showing the chain system $\to$ detector $\to$ amplifier $\to$ observer can be cut at any link with identical statistics [von Neumann 1932]; Bohr elevated the line to a principle while declining to locate it. Bell, sixty years later, made the objection canonical: a "shifty split" is no foundation for a physical theory — nothing in physics should depend on a boundary that is nowhere in particular ["Against 'measurement' ": Bell 1990].

The modern consensus assigns the explanatory work to decoherence: interaction with an environment diagonalizes the reduced density matrix in a preferred pointer basis on absurdly short timescales, and the quantum/classical transition is thereby "explained" [Zeh 1970; Zurek 1981; Zurek 1982; Joos & Zeh 1985; Zurek 2003; Schlosshauer 2004]. The decoherence literature itself is admirably candid that this answers a different question. Diagonality is not an event; a mixture improper for the universe as a whole selects no member of itself; and the program's own reviews state plainly that decoherence does not solve the problem of outcomes [Schlosshauer 2004; Adler 2003]. What decoherence explains is why *interference* becomes unobservable — why the density matrix commutes with the pointer observable — not why *this* pointer position occurs. The cut therefore survives the decoherence program intact: it has merely been renamed the point at which one stops speaking of improper mixtures and starts speaking of occurred events.

This paper claims the cut is physical, locatable, and has a width. The claim is not free-standing: it imports its machinery from a companion paper [Paper 1], which develops outcome selection as a stochastic energy game among detector absorber sites and proves, from stated premises about detectors, that the game is fair and its statistics Born. One theorem of that paper does the work here. **Theorem 2** (restated in §3): an absorber holding energy $e < E_0$ is off-shell by $\Delta E = E_0 - e$ and constitutes a driven, damped *linear* mode with decay rate $\gamma = \Delta E/\hbar$; it is globally contracted onto a fixed point whose phase is slaved to the drive; it possesses no zero mode, no autonomous clock, and no capacity for the phase-coherent feedback that irreversible registration requires. Only within a boundary layer $\Delta E \lesssim \hbar K$ of the shell — fractional width $w = K/\omega$, with K the site–field coupling — does an autonomous phase emerge and locking become possible. Above the layer: dynamics, interference, reversibility. Below it: commitment. The cut is the layer.

Three things must be kept distinct, and the paper's architecture follows the distinction. **Settled, and not ours to claim:** the unitary indifference to the placement of the system–apparatus split (von Neumann's theorem, restated in §6 as the *reason* the conventional cut is movable); decoherence theory in full — einselection, pointer bases, the suppression of interference, with its own conclusion that no outcome is thereby selected; and the demonstration by Allahverdyan, Balian and Nieuwenhuizen that *registration* — the amplification of a selected microscopic asymmetry into a stable macroscopic record — is ordinary statistical mechanics of a metastable apparatus [Allahverdyan et al. 2013]. **Claimed:** that the three-way distinction of §2 (entangling interaction / decoherence / nonlinear lock) is physically real; that the lock supplies exactly the element the settled results lack; that the cut is located at $\gamma \sim K$ with width $w = K/\omega$, computable system by system; and that this yields a detector taxonomy, a reading of the virtual/real distinction, and experimental discriminators. **Open:** whether the sub-threshold slaved-phase statement survives full field quantization (inherited from Paper 1's open problems); the closed dynamical model of the lock itself; and everything involving the entangled sector, which this paper does not touch.

That last point deserves emphasis, because it makes this paper's conditionality strictly weaker than its companion's. Paper 1's premises divide into a single-detector core (wave realism P0; capture energetics P1; amplitude-linear coupling P2; detector initial conditions P3; registration taxonomy P4) and a nonlocal extension (shared registry P5; ordering foliation P6) consumed only by its multi-quantum and entangled-pair constructions. The present paper consumes the core alone. A reader who rejects the nonlocal ontology outright can still evaluate everything here.

The paper proceeds as follows. §2 states the three-way distinction. §3 presents the threshold quantitatively and tabulates the layer width across laboratory systems. §4 maps the anatomy of a measurement event onto the threshold and derives the detector taxonomy. §5 reads the virtual/real distinction as sub-/supra-threshold. §6 explains the movable cut — deriving both its freedom and the boundary of that freedom. §7 states the single-world picture and prices it against the alternatives. §8 assembles the experimental consequences. §9 situates the result and lists open problems.

2. Three things not to conflate

Measurement talk chronically merges three physically distinct processes. Every dispute about the measurement problem, in our reading, is a dispute about the third; clarity about the first two is what the settled literature already provides.

(1) Entangling interaction. Couple system to apparatus and let them interact. On the joint Hilbert space this is plain unitary evolution: $i\hbar\,\partial_t\Psi = H_{\mathrm{tot}}\Psi$. Entanglement grows; the map is linear and *reversible*. Nothing here distinguishes a measurement from any other interaction — which is precisely von Neumann's point, and the origin of the chain.

(2) Decoherence. Open the apparatus to a large environment and trace it out. The reduced state obeys a completely positive, irreversible master equation — but one *still linear in the density matrix*. Off-diagonal terms in the pointer basis are suppressed at spectacular rates; the state comes to *look* like a classical mixture. But linearity is decisive: a linear map cannot send a superposition to one of its branches, only to a mixture of them, weighted exactly as before. Irreversibility has been purchased; selection has not. Dissipation is not nonlinearity.

(3) The lock. The genuinely nonlinear step: a symmetry-breaking dynamics that takes the distributed, decohered excitation and commits it to one basin — one site, one port, one pointer position. This is where an event becomes an event. Paper 1's content is that the lock exists as detector physics, that its statistics are provably Born under stated premises, and that the noise driving it is the ordinary interference of the site's amplitude with the uncontrolled-phase fields at the detector surface. The present paper's content is what the lock's *threshold structure* implies for the geography of the quantum–classical boundary.

In this language the measurement problem decomposes cleanly. Stage (1) is settled unitary dynamics; stage (2) is settled open-system dynamics; the interpretations disagree about stage (3). Everett denies it exists (all basins are occupied); GRW/CSL postulate it as new fundamental stochasticity with new constants [Ghirardi et al. 1986; Bassi & Ghirardi 2003]; Penrose supplies a threshold for it without a mechanism [Penrose 1996]; decoherence-only accounts stop at (2) and redescribe the question. The claim defended here: (3) is ordinary — if nonlinear — detector dynamics, and its threshold is the cut.

The three-way distinction also disciplines vocabulary that this paper will need. *Reversible* applies to stages (1) and, in the echo/rephasing sense, to the capture stage of §4 — evolution whose effects can be undone by feasible operations. *Irreversible-but-linear* is stage (2): information delocalized beyond practical recovery, statistics unchanged. *Committed* is stage (3) alone. The cut, properly speaking, is the boundary of commitment — not of reversibility (decoherence already ends practical reversibility) and not of interaction (everything interacts). Locating the cut therefore means locating the onset of the nonlinearity.

3. The threshold, quantitatively

3.1 The slaved phase and the layer

We restate the companion result in the form used throughout ([Paper 1] Theorem 2 and Appendix B; proofs there). An absorber site short of the full quantum is off-shell: it holds $e < E_0$, with deficit $\Delta E = E_0 - e$, and P4(i) — no stationary state below the transition — makes that deficit a decay rate, $\gamma = \Delta E/\hbar$. In the rotating frame, with amplitude $a = re^{i\varphi}$, detuning Δ, and drive $fe^{i\varphi_d}$:

$$\dot{r} = -\gamma r + f\cos(\varphi_d - \varphi), \qquad \dot{\varphi} = \Delta + \frac{f}{r}\sin(\varphi_d - \varphi).$$

In Cartesian coordinates this system is *linear*, with eigenvalues $\lambda = -\gamma \pm i\Delta$: a unique, globally attracting fixed point, phase offset $\varphi^* = \varphi_d - \arctan(\Delta/\gamma)$ determined by (γ, Δ) alone — identical for identical sites, independent of stored energy. The phase is an enslaved coordinate. No trajectory winds; perturbations decay at rate γ; there are no running solutions and hence no entrainment pull. The site has no clock of its own.

Contrast the autonomous oscillator presupposed by the Adler equation $\dot{\varphi} = \Delta - K\sin\varphi$ [Adler 1946]: amplitude held on a limit cycle by the oscillator's own gain, phase a neutrally stable zero mode, free to run — and, when running, exerting the nonzero average pull that is the engine of injection locking, laser mode competition, and every rich-get-richer instability in synchronization physics [Pikovsky et al. 2001]. The qualitative difference between the two regimes is the theorem: *below threshold there is no zero mode to entrain.* A free clock — the prerequisite for systematic, phase-coherent energy extraction, and hence for winner-take-all feedback and for irreversible commitment — emerges only where the restoring rate collapses, $\gamma \lesssim K$, i.e. within a boundary layer

$$\Delta E \lesssim \hbar K \qquad \Longleftrightarrow \qquad w \equiv \frac{\Delta E_{\text{layer}}}{E_0} = \frac{K}{\omega}$$

of the shell. Figure 1(a) displays the transition as a single observable: an injected oscillator's mean phase-winding rate, swept through threshold. Below, exactly zero — the drive owns the phase. Above, the free clock switches on and runs at the autonomous Adler rate. The crossover is the layer.

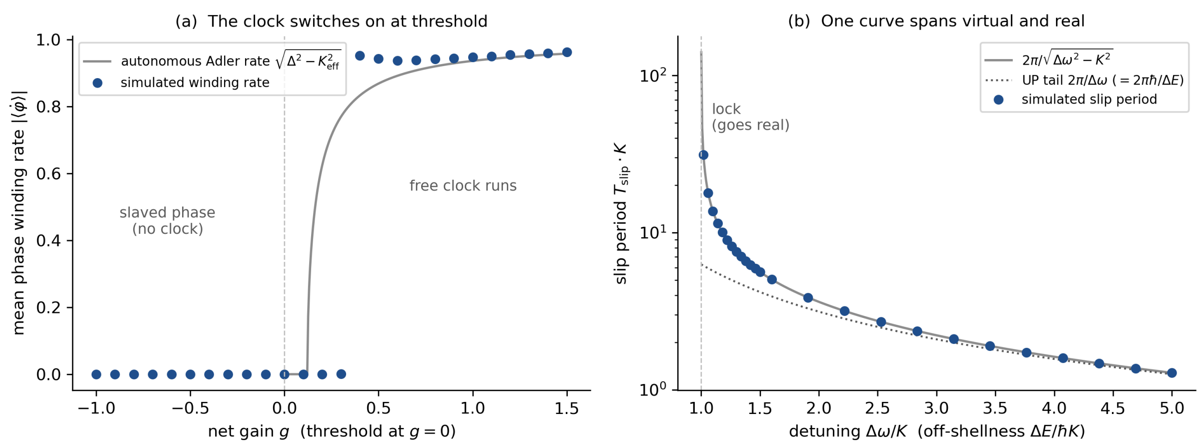

Figure 1: **Figure 1.** The layer as physics. **(a)** Injected Stuart–Landau oscillator, net gain g swept through threshold at fixed detuning and injection: mean phase-winding rate $|\langle\dot{\varphi}\rangle|$ is identically zero in the slaved (linear, sub-threshold) regime and switches on above threshold, following the autonomous Adler rate $\sqrt{\Delta^2 - K_{\text{eff}}^2}$ once the injection can no longer entrain the free clock. **(b)** One curve spans virtual and real (§5): Adler slip period versus detuning. Far off-resonance the slip period is the energy–time-uncertainty lifetime $2\pi\hbar/\Delta E$ (dotted); at the tongue boundary it diverges — critical slowing — and the mode locks: goes on-shell. Simulated points; analytic curves; methods in Appendix A.

3.2 The width, two ways

Two independent estimates give the same layer width ([Paper 1] Appendix B.2). *Marginality:* the phase becomes exploitable when its restoring rate falls below the coupling, $\gamma \lesssim K$, giving $w = K/\omega$ directly. *Width–commitment identity:* the linewidth Γ that broadens the final state is, by fluctuation–dissipation, also the rate of irreversible decay into registered products; with $K \sim \Gamma$ this reproduces $w = \Gamma/\omega$. The agreement is not a coincidence but the same physics read from the dynamical and the dissipative side — and it is the reason the layer width is an *empirically tabulated* quantity: it is a ratio of a linewidth to a transition frequency, both measured.

3.3 The cut has an address: the layer width across laboratory systems

The following are order-of-magnitude estimates assembled from standard linewidth and frequency scales; each entry is to be sourced precisely in the citation pass [values to verify before circulation].

System	phase-restoring / linewidth scale Γ (s^{-1})	transition frequency ω (s^{-1})	$w = \Gamma/\omega$	character of the cut
Atomic line (alkali D-line, natural width)	10^7–10^8	$\sim 2 \times 10^{15}$	10^{-8}–10^{-6}	razor-sharp; Born exact to all practical resolution
Trapped-ion optical clock transition	$\lesssim 10^1$	$\sim 3 \times 10^{15}$	$\lesssim 10^{-14}$	sharpest cut in any laboratory
Superconducting qubit ($T_{1,2} \sim 10$–100 μs)	10^4–10^5	$\sim 3 \times 10^{10}$	10^{-6}–10^{-5}	atomic-grade sharpness in a macroscopic circuit
Semiconductor quantum dot (4 K)	10^9–10^{11}	$\sim 2 \times 10^{15}$	10^{-6}–10^{-4}	intermediate; the natural dial (§8.2)
Broadband photodiode / SPAD (300 K)	10^{13}–10^{14}	$\sim 3 \times 10^{15}$	10^{-2}	percent-level layer; deviations live here ([Paper 1] §8)

Two readings of the table. First, the cut is *not* at a universal scale: it sits at a different absolute energy in every system, exactly as a convention-based cut would — but its *location in the dimensionless variable* γ/K is universal, and its width is predicted, not chosen. Second, the table dissolves a familiar puzzle: superconducting circuits are "macroscopic" by any particle-count measure, yet they interfere, violate Bell and Leggett–Garg inequalities, and behave as clean two-level systems. On the threshold account this is unmysterious: their w is atomic-grade. Macroscopicity in mass or particle number is simply not the variable; γ/K is. §8.4 turns this observation into a discriminator.

4. Anatomy of a measurement event

4.1 Three stages, one crossing

Paper 1 decomposes a measurement event into capture (the incident wave reversibly excites the detector), selection (one site ends up with the quantum), and registration (the outcome becomes irreversible and macroscopic). The cut, as defined here, is crossed during *selection*, at the lock — and the three-stage structure gives the crossing a time-domain profile ([Paper 1] §6.1): capture completes on the fastest timescale the interaction permits ($\tau \gtrsim \hbar/\Delta E$, saturated at $\sim 1/\omega$ for a massless quantum); the lock engages reversibly on the fast synchronization scale; irreversibility is supplied only by the slow dissipative commit at the bottom of the ladder, where the assembled quantum is delivered to the bath and the record is written.

The stage boundaries align with the vocabulary of §2. Capture is stage-(1) physics, and demonstrably reversible: spin echoes, photon echoes, and atomic-frequency-comb quantum memories run it backwards on the bench, at the single-photon level, with no registration having occurred [Hahn 1950; Kurnit et al. 1964; de Riedmatten et al. 2008; Afzelius et al. 2009]. Selection is stage-(3) physics — the fair game of Paper 1 — and it is here, precisely at the locking layer, that the system leaves the reversible sector. Registration is stage-(2)-plus-amplification: the Curie–Weiss physics of Allahverdyan–Balian–Nieuwenhuizen, importing the selected asymmetry and making it permanent [Allahverdyan et al. 2013]. The commonplace that "the collapse happens when the record is amplified" misassigns the cut by one stage: amplification finalizes; the lock decides. Paper 1's Theorem 4 (commit-rate independence) is the quantitative form of the distinction — the speed of the slow finalization drops out of the outcome statistics entirely.

4.2 The detector taxonomy as threshold phenomenology

Two classifications of real detectors follow from the threshold structure, and their cross-product is exhaustive.

By spectrum (P4). *Discrete-level absorbers* (atoms, molecules, ions): no stationary state below the transition, so the full quantum must assemble at one site; the layer is razor-thin (Table, rows 1–2); sub-threshold excitation is virtual and unbankable — level discreteness enforces threshold sharpness. *Continuum absorbers* (semiconductors, metals): final states at all energies above the gap; registration is a rate process throughout selection; the layer is wide ($w \sim 10^{-2}$) and the percent-level deviation phenomenology of Paper 1 §8 lives in it.

By record structure. *Event detectors* (photomultiplier, SPAD, single atom): one lock, one record — the selection game runs once. *Track-recording media* (cloud chamber, emulsion, silicon tracker): Mott's problem [Mott 1929] — a single spherical wave produces a straight track — is, in threshold language, a *sequence of partial locks*: each grain along the track runs a selection game whose initial stakes are conditioned by the preceding lock (the wave re-formed around the previous registration), so the sequence is a chain of conditional Born games, each crossing its own layer. Mott's unitary demonstration that the amplitude concentrates on collinear configurations supplies the conditional stakes; the locks supply the events. A track is not one measurement but a correlated cascade of them — which is why track detectors, not interferometers, are where the classical world's *trajectories* come from.

The taxonomy earns its keep twice below: it sorts every entry of the deviation phenomenology (§8 inherits Paper 1's rule that class-(ii) continuum detectors evade late-game bias entirely via commit-rate independence), and it blocks a familiar false generalization — conclusions drawn from event detectors do not automatically transfer to tracking media, whose cascade structure is the reason "watching a trajectory" feels classical.

5. Virtual and real: the threshold as the shell

The same threshold that draws the cut draws the virtual/real line, and the identification is quantitative, not metaphorical. The dictionary: off-shell energy imbalance ΔE corresponds to detuning $\hbar\Delta\omega$; the lock threshold is the mass shell.

A virtual excitation is an unlocked, off-resonance mode. It slips against the drive and dies as a transient; its lifetime is the slip time of a failed lock. For the Adler phase outside the locking range, the slip period is

$$\tau_{\text{slip}} = \frac{2\pi}{\sqrt{\Delta\omega^2 - K^2}} \xrightarrow{\Delta\omega \gg K} \frac{2\pi}{\Delta\omega} = \frac{2\pi\hbar}{\Delta E},$$

which *is* the energy–time uncertainty lifetime — not postulated but produced, as the beat period of an excitation that fails to lock (Figure 1(b), dotted asymptote). At the tongue boundary $\Delta\omega \to K^+$ the slip period diverges: critical slowing. The excitation stops slipping, locks, and persists — it has gone on-shell. One curve spans both regimes (Figure 1(b)): the far tail is the uncertainty-principle lifetime of the virtual; the divergence is the escape to reality; the threshold between them is the same $\gamma \sim K$ layer that §3 identified as the cut. The governing clock changes at the crossing: $\hbar/\Delta E$ (off-shell borrowing) below; infinite (stable attractor) or $\hbar/\Gamma$ (golden-rule decay through open channels) above. For a real but unstable particle the product $\Gamma\tau \sim \hbar$ still *resembles* the uncertainty principle, but as a consequence — the Fourier width of an exponentially decaying amplitude — not as the off-shell borrowing that governs the virtual side.

Status of this section, stated plainly. The identification is a *re-description with one quantitative consequence*, not an independent derivation: standard quantum field theory computes with off-shell propagators and needs no story about what a virtual excitation "is doing," and "virtual-particle lifetime" is itself a heuristic gloss on a propagator integral. What the re-description adds is the middle of Figure 1(b): a definite, measurable crossover profile — the critical-slowing divergence at $\Delta E \to \hbar K$ — where the standard heuristic offers only the two asymptotes. Systems that can be tuned through the layer (§8.2) can in principle trace the middle of the curve; that, and not the asymptotes, is the falsifiable content. We flag the section's altitude accordingly and rest no later result on it.

6. The movable cut, explained

6.1 Why the conventional cut is movable: a theorem, not an embarrassment

Within the reversible sector, the placement of the system–apparatus split is provably free. Von Neumann's chain argument is exactly this: model the detector quantum-mechanically or classically, cut after the system, after the detector, after the amplifier — all choices yield identical statistics, because the intervening dynamics is linear and the eventual outcome weights are basis-quadratic either way [von Neumann 1932]. The threshold account *keeps this theorem and explains its domain.* Every link of the chain that lies above the layer — every stage whose dynamics is stage-(1) unitary or stage-(2) linear-dissipative — can be placed on either side of the description at zero cost, precisely because linear dynamics commutes with the bookkeeping. The freedom to move the cut is a symptom of linearity, and linearity is exactly what holds *everywhere except the layer.*

The freedom ends where the nonlinearity begins. The lock cannot be placed on the "described unitarily" side of the split, because no unitary (or linear-dissipative) description reproduces commitment: a linear map preserves the superposition's weights and can only deliver mixtures (§2). The layer is therefore the unique link of the chain at which the two descriptions cease to be intertranslatable — the one place the cut *must* go if the description on each side is to be dynamically accurate. The conventional cut's shiftiness and the physical cut's definiteness coexist without tension: the former is the linearity theorem, the latter is the locking threshold, and the confusion between them dissolves once stages (1)–(3) are kept separate.

Bell's complaint is thereby answered in its own terms. The split is shifty *within the sector where nothing depends on it* — that is a theorem, and no foundation should be built on resisting a theorem. It is pinned — located at $\gamma \sim K$, with width $w = K/\omega$ — exactly where something *does* depend on it. Nothing fundamental is nowhere in particular.

6.2 The cut as a crossover

A boundary with a dimensionless location and a finite width is, in the language of many-body physics, a *crossover*, and the naming buys discipline. The location is $\gamma/K = 1$; the width is $w = K/\omega$; on one side one set of laws dominates (reversible, interference-preserving, split-placement-free), on the other a different term dominates (dissipative, committing, split-pinning). Nothing discontinuous happens to the laws themselves — the locking term is present in the dynamics throughout, and simply has nothing to grip while the phase is slaved. A crossover, not a phase transition; a dominance boundary, not a change of legislation.

Two consequences of the naming. First, *dose–response*: observables that probe the boundary should vary smoothly with the crossover variable γ/K, with all deviation phenomenology scaling with the layer width — which is exactly the structure of Paper 1's deviation ledger (deviations $O(w)$: negligible for atomic detectors, percent-level for broadband solids). A conventional cut predicts no such systematics; a crossover mandates them. Second, *completion elsewhere*: in the parent framework the dissipative sector is also where a preferred foliation enters the dynamics, so the same crossover marks where frame-sensitive physics switches from present-but-decoupled to dynamically active. That identification, its symmetry accounting, and its experimental program belong to the framework paper and are deliberately not developed here; we note only that the cut's crossover structure is what makes such a completion well-formed [Paper 3, in preparation].

7. Single world: the occupied basin

The threshold picture yields a one-world reading of the formalism. Above the layer, the wave explores; superposition is physical, interference is real, and *no basin is occupied* — the selection game has not run. At the layer, symmetry breaks: noise — the surface-interference noise identified in Paper 1 — tips the distributed excitation into one basin, and the system *occupies* it. The other basins do not become other worlds; they become unoccupied attractors, exactly as the untaken well of a buckling beam is not a parallel beam. Where Everett reads the persistence of the other branches' amplitudes as their reality, the threshold account reads their amplitudes as *stakes that lost* — physically real before the game (P0 wave realism is not negotiable here), physically redistributed by it (the energy went to the winner; Paper 1 §4).

Pricing against the alternatives, with our own costs on the table:

Program	supplies an event?	mechanism?	new constants?	principal cost
Decoherence-only	no	—	no	redescribes the question; cut survives renamed
Everett/MWI	no (all outcomes)	—	no	all basins real; Born weights need decision theory
GRW/CSL	yes	postulated noise	yes (λ, r_C)	new physics engineered for the job; anomalies predicted [Bassi & Ghirardi 2003; Bassi et al. 2013]
Penrose OR	threshold only	none offered	gravitational scale	location without dynamics [Penrose 1996]
ABN (Curie–Weiss)	registration only	statistical mechanics	no	selection assumed at the crucial step [Allahverdyan et al. 2013]
This program	yes	detector locking (Paper 1)	no	conditional on P0–P4; lock dynamics not yet closed microscopically

The comparison is deliberately double-edged. Against GRW/CSL: no new constants, no engineered noise — the fairness is proved, not tuned, and the "collapse rate" is the physical commit rate of an actual detector. Against Penrose: a mechanism where he supplies only a scale — though his structural instinct, that the boundary is physical and dynamical rather than conventional, is the same instinct this paper formalizes. Against MWI: one world, at the price of wave realism and the two open problems we carry visibly (the field-quantized sub-threshold statement; the closed lock dynamics). Against decoherence-only: we accept every einselection result wholesale and add the one thing its own reviews concede is missing. The honest summary is that every program pays somewhere; this one pays in premises about detectors, which have the redeeming feature of being checkable in laboratories rather than in metaphysics.

8. Where the cut bites: experimental consequences

8.1 Inherited: the deviation phenomenology scales with the layer

Everything in Paper 1's deviation ledger is, in the present language, layer phenomenology: deviations from exact Born statistics are $O(w)$, hence invisible for atomic-grade detectors and percent-level where $w \sim 10^{-2}$ (broadband solids), concentrated in configurations that couple to the layer (mismatched ports; pre-cohered absorbers). We do not re-derive that ledger; we re-read it as the dose–response curve of the crossover (§6.2): *the same experiments that test Paper 1's selection mechanism test this paper's cut location*, because the deviations are the layer made visible.

8.2 The dial: systems engineered to straddle the layer

The table of §3.3 spans eight orders of magnitude in w, and — decisively — some platforms can *tune* it. A semiconductor quantum dot's linewidth varies orders of magnitude with temperature and phonon coupling; a superconducting qubit's γ is engineered via its Purcell environment; cavity QED sets K directly by detuning and geometry. The threshold account makes a definite prediction with no analogue in convention-based accounts: *outcome statistics, lock-time distributions, and interference-recovery windows should vary systematically with the dimensionless ratio γ/K — and with nothing else.* Two systems with identical γ/K and wildly different mass, size, particle number, or temperature should show identical boundary phenomenology; two systems differing only in γ/K should trace the crossover profile of Figure 1. This is cheap to state and, on platforms that already sweep these parameters for engineering reasons, plausibly cheap to test; we flag it as the paper's primary novel experimental claim.

8.3 Macromolecule interferometry, re-read

The interference of large molecules — C_{70}, functionalized oligomers, and by 2019 a 25 kDa, 2000-atom molecule [Arndt et al. 1999; Hornberger et al. 2012; Fein et al. 2019] — is standardly narrated as a march up the mass axis toward an anticipated breakdown. The threshold account predicts the march continues indefinitely *as long as the interfering system stays above its layer*: mass is not the variable. Visibility is lost to decoherence (thermal emission, collisional which-path labeling — stage-(2) physics, fully accounted by the standard theory [Hornberger et al. 2012]) and *committed* only when a detector-grade lock occurs. The discriminating structure is recoverability: stage-(2) visibility loss is in-principle reversible (echo-type recovery; capture-without-registration, §4.1), stage-(3) loss is not. An experiment that could rephase after a putative "collapse" — recovering visibility that CSL or gravitational reduction says is gone for good — separates the programs cleanly: spontaneous-collapse theories predict a mass-scaled, unrecoverable loss channel; the threshold account predicts zero unrecoverable loss until coupling to an above-threshold system, at any mass.

8.4 Leggett–Garg tests as cut-location experiments

Macrorealism, as operationalized by the Leggett–Garg inequality [Leggett & Garg 1985; Emary et al. 2014], asks whether a system has a definite state at all times. The threshold account's answer is sharp: definiteness is *purchased at the layer*, so LGI violations should persist for any system maintained below threshold ($\gamma \ll K$… i.e. kept coherent, weakly monitored) regardless of macroscopicity, and disappear exactly when the monitoring chain includes an above-threshold lock. Superconducting circuits — macroscopic by particle count, atomic-grade by w (Table §3.3) — are the existing confirmation: they violate LGI robustly. The discriminator against mass-scaled programs recurs: CSL predicts LGI violations fail at a mass/size scale independent of coupling; the threshold predicts they fail at a coupling scale independent of mass. The two axes are experimentally orthogonal, and platforms exist along both.

8.5 What would falsify this paper

(i) A system demonstrably held above its layer ($\gamma \ll K$, autonomous phase confirmed by injection-locking phenomenology) that nonetheless shows *unrecoverable* interference loss scaling with mass or size — the CSL/OR signature — would falsify the claim that γ/K is the only boundary variable. (ii) A demonstration that outcome commitment occurs in a regime where the phase is provably slaved (e.g. registration without any mode crossing its layer) would falsify the identification of the cut with the lock. (iii) Boundary phenomenology (deviation statistics, lock-time distributions) that fails to collapse onto the single variable γ/K across platforms would falsify the crossover structure. (iv) Inherited from Paper 1: a controlled open-system calculation showing the slaved-phase statement fails under field quantization would remove this paper's foundation — we flag it as the live theoretical risk, not a rhetorical one.

9. Discussion

Relation to the settled literature. This paper is structured as a completion, not a rival, of three bodies of work. Decoherence theory supplies stage (2) entire; we adopt its pointer bases, its timescales, and its own verdict that selection lies outside its scope. Allahverdyan–Balian–Nieuwenhuizen supply registration; their Curie–Weiss apparatus is, in our language, the machinery *below* the layer, and their explicit assumption of a selected asymmetry is the interface our stage (3) fills. Von Neumann's movability theorem is preserved with its domain finally drawn. The genuinely contested relationship is with the programs that also claim stage (3): against engineered-noise reduction models we claim the same phenomenological structure with the noise identified and the fairness proved rather than stipulated (Paper 1), at the price of conditionality on detector premises; against gravitational reduction we claim a mechanism and a non-gravitational, non-universal threshold — the cut is at $\gamma \sim K$, which gravity does not set.

Open problems. (i) The field-quantized sub-threshold statement (inherited; the decisive one). (ii) The closed dynamical model of the lock — Paper 1's surface-interference sketch grounds the noise's form; the conserving nonlinear dynamics remains open. (iii) The quantitative crossover profile: Figure 1 is a minimal model; a first-principles calculation of the winding-rate and slip-time profiles for a physical absorber (with the P4(i) level structure explicit) would convert §8.2's dial prediction from structural to numerical. (iv) The track-cascade statistics of §4.2: Mott's conditional stakes composed with per-grain Born games should reproduce the full track-statistics phenomenology (straggling, delta rays aside); a worked calculation would make the taxonomy's second row as quantitative as its first. (v) The crossover's completion in the framework paper [Paper 3].

The one-sentence summary. The Heisenberg cut is the locking layer of the detector: free to place wherever linearity reigns — which is a theorem — and pinned, with computable width $w = K/\omega$, exactly where it does not; the classical world begins where the phase stops being slaved, and that address is different, calculable, and testable in every laboratory in physics.

Appendix A — Figure methods

Panel (a): injected Stuart–Landau oscillator $\dot{a} = (g - |a|^2)a - i\Delta a + F + \xi(t)$, $\Delta = 1.0$, $F = 0.35$, complex white noise of strength $D = 10^{-4}$; Euler–Maruyama, $dt = 0.01$; transient $T = 200$ discarded, winding accumulated over $T = 800$; 26 gain values $g \in [-1, 1.5]$; overlay $\sqrt{\Delta^2 - F^2/g}$ for $g > (F/\Delta)^2$ (limit-cycle amplitude $\sqrt{g}$, effective locking range $K_{\text{eff}} = F/\sqrt{g}$). Below threshold the measured winding rate is $\lesssim 4 \times 10^{-5}$ (noise floor). Panel (b): Adler equation $\dot{\varphi} = \Delta\omega - K \sin\varphi$, $K = 1$; slip period measured as the time for one full 2π advance, $dt = 5 \times 10^{-4}$; 25 detunings $\Delta\omega/K \in [1.02, 5]$; analytic curve $2\pi/\sqrt{\Delta\omega^2 - K^2}$; at $\Delta\omega/K = 5$ simulation and analytic agree to 4×10^{-4} relative. Script: `paper2_sims/threshold_clock.py`, seed pinned (20260728); output `figures/p2_fig1_threshold.png`.

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(Author–year style, alphabetical. Entries marked [P1-verified] were bibliographically verified during the companion paper's citation passes; all remaining entries were verified against publisher/indexer records on 2026-07-28 — the verification log, with per-entry source URLs, is `PAPER2_citation_verification_2026-07-28.md` alongside this draft.)

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